

Discrete Mathematics and Graph Theory

Module 3 (CO3) — Graph Theory

Questions with Detailed Step-by-Step Solutions

Every question is solved from zero to the final answer — what is used, why it is used, and every small step explained. Each solution begins with what the question asks, then the idea or formula used and why, then the complete step-by-step working, and finally the answer. Part A questions carry 2 marks; Part B questions carry 7 marks.

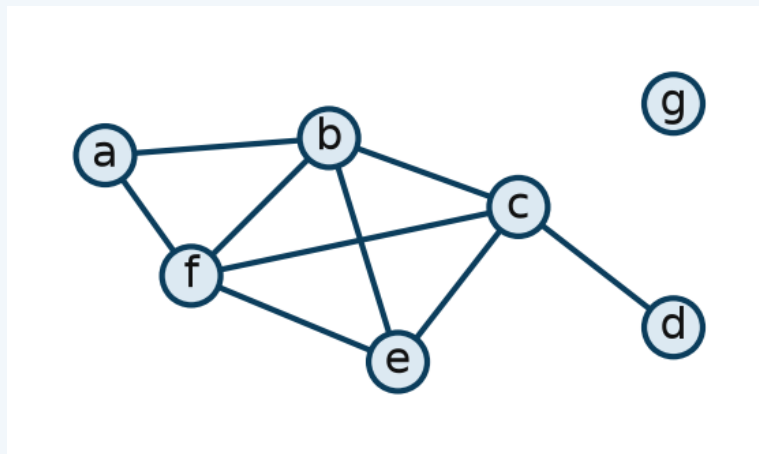
Part A — Short Questions (2 Marks)

Q1. (2 marks) Define the degree of a vertex in a graph. How does a loop affect the degree?

Solution — full explanation.

What the question asks: define “degree” and state the special rule for loops. **Definition.** The **degree** of a vertex v , written $\deg(v)$, is the number of edges incident with v (i.e. edges having v as an endpoint). **The loop rule.** A loop is an edge from a vertex to itself. Its two ends are both at the same vertex, so a **loop contributes 2** to the degree of that vertex. **Example.** For the graph below:

$$\deg(a) = 2, \deg(b) = 4, \deg(c) = 4, \deg(d) = 1, \deg(e) = 3, \deg(f) = 4, \deg(g) = 0.$$



Why the loop rule? Every edge must contribute exactly two “edge-ends” (one at each endpoint). A loop’s two ends sit on the same vertex, so it counts twice there. **Final answer.** The degree of v is the number of edges incident with it, with a loop counting 2. Degree 0 = isolated vertex; degree 1 = pendant (leaf) vertex.

Q2. (2 marks) State the Handshaking theorem. **Solution — full explanation.**

What the question asks: state the basic relation between the degrees of the vertices and the number of edges. **Statement.** In any undirected graph $G = (V, E)$ with e edges,

$$2e = \sum_{v \in V} \deg(v).$$

Meaning. The sum of the degrees of all the vertices equals **twice** the number of edges. **Why it is true.** Each edge has two endpoints, so it adds 1 to the degree of each endpoint — that is 2 to the total degree-sum. Summing over all edges counts every edge twice, hence the total is $2e$. **Two consequences.** (i) The degree-sum is always even. (ii) The number of vertices of **odd** degree is always even. **Final answer.** $2e = \sum_{v \in V} \deg(v)$ — the sum of the degrees is twice the number of

edges.

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Q3. (2 marks) A graph contains 21 edges, 3 vertices of degree 4 and all other vertices of degree 2. Calculate the total number of vertices. **Solution — full explanation.**

What the question asks: find the total number of vertices n . **Which idea is used and why.** The Handshaking theorem links the degree-sum to the number of edges: $2e = \sum \deg(v)$. Since e is given and the degree pattern is known, we can solve for n . **Set up.** Let n = total vertices. Then 3 vertices have degree 4 and the remaining $(n - 3)$ vertices have degree 2. Also $e = 21$. **Apply the theorem.**

$$3(4) + (n - 3)(2) = 2(21) \Rightarrow 12 + 2n - 6 = 42 \Rightarrow 2n + 6 = 42 \Rightarrow 2n = 36 \Rightarrow n = 18.$$

Check. Degrees: 3 vertices of degree 4 and 15 vertices of degree 2 give $12 + 30 = 42 = 2 \times 21$.

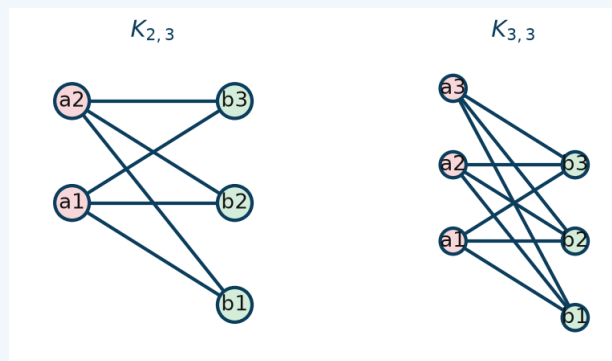
✓**Final answer.** The total number of vertices is 18.

Q4. (2 marks) How many edges does $K_{2,6}$ contain? **Solution — full explanation.**

What the question asks: count the edges of the complete bipartite graph $K_{2,6}$. **Definition recap.** $K_{m,n}$ has two parts of sizes m and n ; every vertex of the first part is joined to every vertex of the second part, and there are no edges within a part. **Formula and why.** Each of the m vertices in part 1 connects to all n vertices in part 2, giving $m \times n$ edges. So

$$|E(K_{m,n})| = m \times n.$$

Compute. $|E(K_{2,6})| = 2 \times 6 = 12$.



Final answer. $K_{2,6}$ contains 12 edges.

Q5. (2 marks) How many edges does $K_{3,6}$ contain? **Solution — full explanation.**

What the question asks: count the edges of the complete bipartite graph $K_{3,6}$. **Idea.** Same formula as Q4: $|E(K_{m,n})| = m \times n$, because every one of the m vertices in the first part joins to all n in the second. **Compute.** $|E(K_{3,6})| = 3 \times 6 = 18$. **Final answer.** $K_{3,6}$ contains 18 edges.

Q6. (2 marks) How many edges does K_{100} contain? **Solution — full explanation.**

What the question asks: count the edges of the complete graph on 100 vertices. **Formula and why.** K_n has one edge between every pair of distinct vertices. The number of pairs is the number of ways of choosing 2 vertices out of n :

$$|E(K_n)| = \binom{n}{2} = \frac{n(n-1)}{2}.$$

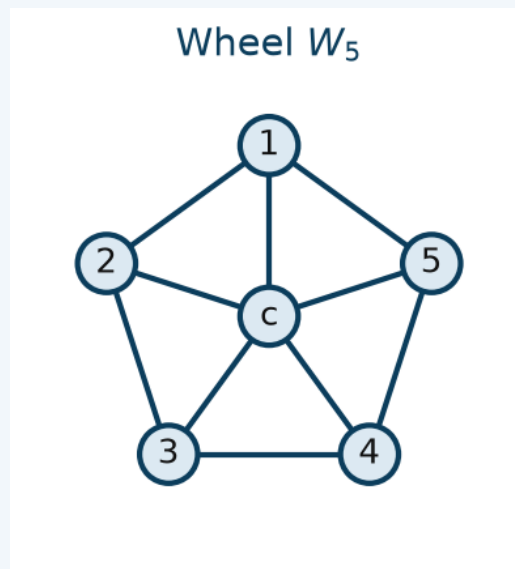
Compute. $|E(K_{100})| = \frac{100 \times 99}{2} = 50 \times 99 = 4950$. **Final answer.** K_{100} contains **4950** edges.

Q7. (2 marks) How many edges does W_{100} contain? **Solution — full explanation.**

What the question asks: count the edges of the wheel W_{100} . **Structure of W_n .** A wheel is a cycle C_n (the rim) plus one extra central vertex (the hub) joined to all n rim vertices (the spokes).

Count and why. Rim edges: n . Spokes: n . So the total is

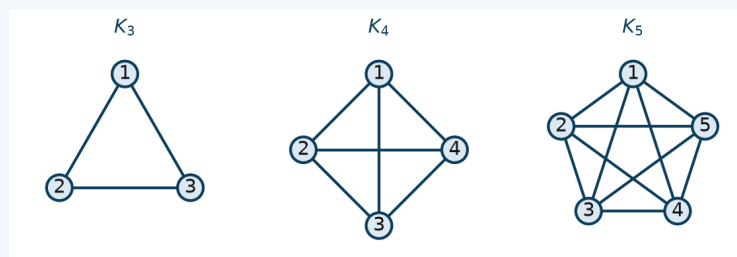
$$|E(W_n)| = n + n = 2n.$$



Compute. $|E(W_{100})| = 2 \times 100 = 200$. **Final answer.** W_{100} contains **200** edges.

Q8. (2 marks) Find the chromatic number of the complete graph K_5 . **Solution — full explanation.**

What the question asks: find the minimum number of colours needed to colour K_5 so that no two adjacent vertices share a colour. **Idea and why.** In K_n every pair of vertices is adjacent. So every vertex is adjacent to all the others, and therefore each vertex needs a colour different from all the rest — we need n distinct colours. Hence $\chi(K_n) = n$. **Apply.** $\chi(K_5) = 5$.



Final answer. The chromatic number of K_5 is **5**.

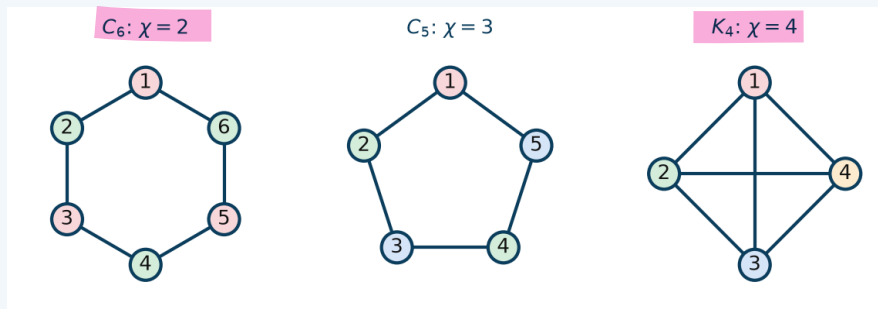
Q9. (2 marks) What is the chromatic number of an odd cycle? **Solution — full explanation.**

What the question asks: find the minimum number of colours for a cycle C_n with n odd. **Idea and why.** An even cycle can be 2-coloured (colour alternately), but an odd cycle cannot: after colouring alternately all the way round, the last edge would join two vertices of the same colour.

Since a graph is 2-colourable if and only if it is bipartite (no odd cycle), an odd cycle needs at least 3 colours. And 3 are enough (colour one vertex a third colour). Hence

$$\chi(C_n) = 3 \quad \text{for odd } n.$$

(C6) = a cycle with 6 vertices.
(K4) = complete graph with 4 vertices



Final answer. The chromatic number of an odd cycle is 3.

Q10. (2 marks) What is the chromatic number of the complete graph K_4 ? **Solution — full explanation.**

What the question asks: minimum colours to colour K_4 . **Idea.** As in Q8, every vertex of K_n is adjacent to all the others, so all n colours are required: $\chi(K_n) = n$. **Apply.** $\chi(K_4) = 4$. **Final answer.** The chromatic number of K_4 is 4.

Q11. (2 marks) Write Euler's formula for a planar graph with 4 components. **Solution — full explanation.**

What the question asks: give the vertex–edge–region relation for a planar graph that has 4 components. **The formula and why.** For a *connected* planar graph, Euler's formula is $v - e + r = 2$. If the planar graph has k components, the formula generalises to

$$v - e + r = k + 1,$$

because each extra component adds one more region (the “outside” of the whole picture is counted once, and each separate piece contributes one extra face). **Apply with** $k = 4$.

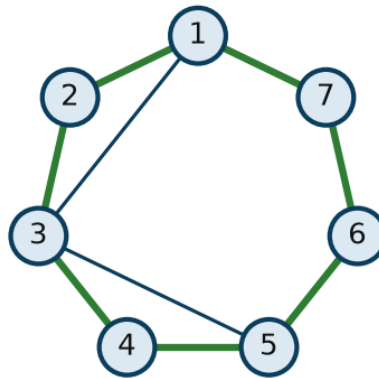
$$v - e + r = 4 + 1 = 5 \implies v - e + r = 5.$$

Final answer. For a planar graph with 4 components: $v - e + r = 5$.

Q12. (2 marks) Define a Hamiltonian circuit. **Solution — full explanation.**

What the question asks: give the definition of a Hamiltonian circuit. **Definition.** A **Hamiltonian circuit** is a cycle in a connected graph that contains *all* the vertices of the graph — each vertex is visited exactly once, except the starting vertex, which is also the ending vertex. Equivalently, it is a closed Hamiltonian path. **Note.** A graph that contains a Hamiltonian circuit is called a **Hamiltonian graph**. Removing any one edge of a Hamiltonian circuit gives a Hamiltonian path.

Hamiltonian circuit (green)

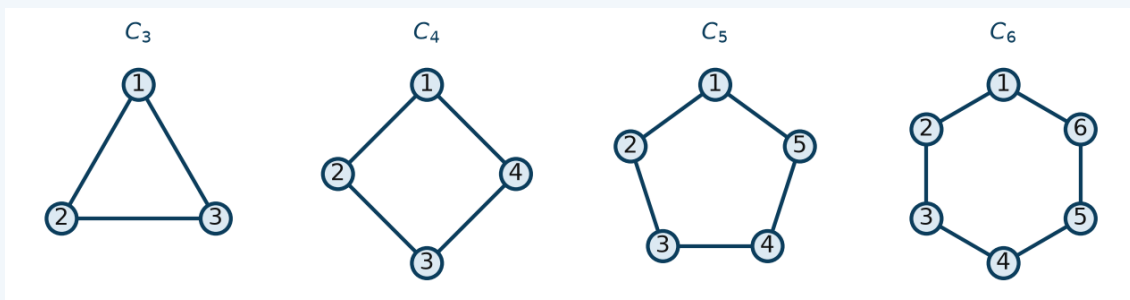


Final answer. A Hamiltonian circuit is a cycle visiting every vertex exactly once and returning to the start.

Q13. (2 marks) Determine which of the following graphs is bipartite: (a) C_3 (b) C_5 (c) C_4 (d) C_6 . **Solution — full explanation.**

What the question asks: decide which of these cycles can be split into two independent parts. **Key fact and why.** A graph is bipartite if and only if it has **no odd cycle**. So an even cycle is bipartite and an odd cycle is not. **Apply.**

- (a) C_3 — length 3 (odd) $\Rightarrow$ **not bipartite**.
- (b) C_5 — length 5 (odd) $\Rightarrow$ **not bipartite**.
- (c) C_4 — length 4 (even) $\Rightarrow$ **bipartite**. ✓
- (d) C_6 — length 6 (even) $\Rightarrow$ **bipartite**. ✓



Final answer. The bipartite graphs are C_4 and C_6 .

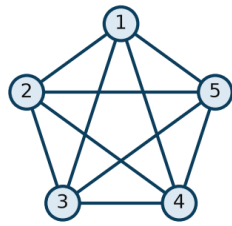
Q14. (2 marks) Show that K_5 is not planar. **Solution — full explanation.**

What the question asks: prove that K_5 cannot be drawn in the plane without edge crossings. **Tool and why.** For a connected simple planar graph with $v \geq 3$ vertices and e edges, the bound $e \leq 3v - 6$ must hold (it comes from Euler's formula). If K_5 violates this bound, it cannot be planar. **Compute for K_5 .** Here $v = 5$ and $e = \binom{5}{2} = 10$. The bound gives

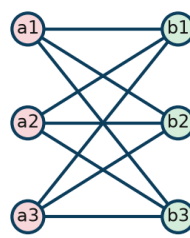
$$3v - 6 = 3(5) - 6 = 9.$$

But $e = 10 > 9$. **Conclusion.** Since $10 > 9$, K_5 violates $e \leq 3v - 6$, so K_5 is **not planar**.

K_5 : not planar



$K_{3,3}$: not planar



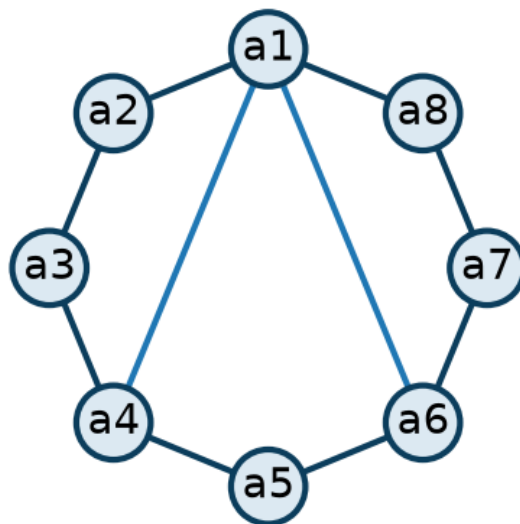
Final answer. K_5 is not planar (its 10 edges exceed the planar maximum of 9 for 5 vertices).

Part B — Long Questions (7 Marks)

Q15. (7 marks) Define a bipartite graph. Check whether the following graph G is bipartite or not. **Solution — full explanation.**

What the question asks: first give the definition, then test the given 8-vertex graph. **Definition.** A simple graph G is **bipartite** if its vertex set V can be split into two disjoint sets V_1 and V_2 such that *every* edge joins a vertex of V_1 to a vertex of V_2 (no edge has both ends in the same part). **Test used and why.** A graph is bipartite iff it has no odd cycle. Equivalently, we try a 2-colouring: colour a vertex red, all its neighbours green, all neighbours of green vertices red, and so on. If we finish with no conflict, the graph is bipartite.

G (8 vertices)



Apply to the given graph. Take

$$V_1 = \{a_1, a_3, a_5, a_7\}, \quad V_2 = \{a_2, a_4, a_6, a_8\}.$$

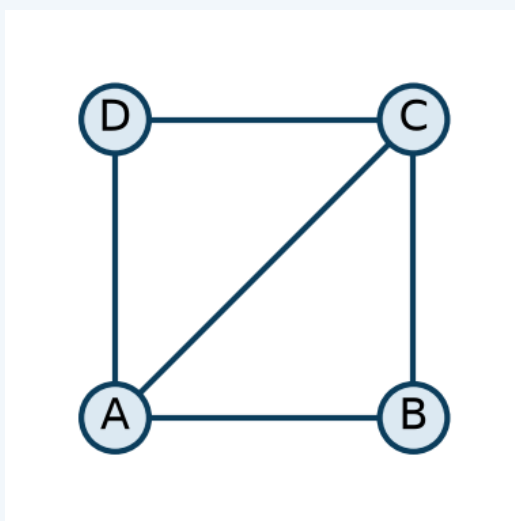
Check the edges:

- The cycle edges $a_1a_2, a_2a_3, \dots, a_8a_1$ each join a vertex of V_1 to one of V_2 . ✓
- The chords a_1a_4 and a_1a_6 : here $a_1 \in V_1$ and $a_4, a_6 \in V_2$. ✓

Every edge joins V_1 to V_2 , so the 2-colouring works. **Final answer.** Yes — the graph G is bipartite, with parts $\{a_1, a_3, a_5, a_7\}$ and $\{a_2, a_4, a_6, a_8\}$.

Q16. (7 marks) Define an Euler circuit. State the necessary and sufficient condition for a connected graph to have an Euler circuit. Check whether the following graph has an Euler circuit or not. **Solution — full explanation.**

What the question asks: define an Euler circuit, state the test, then apply it to the given graph. **Definition.** An Euler circuit is a simple circuit that contains every edge of the graph exactly once and returns to the starting vertex. **Condition and why.** A connected graph has an Euler circuit if and only if every vertex has even degree. Reason: each time the walk passes through a vertex it uses one edge to arrive and one to leave, so edges are used in pairs — every degree must be even.



Apply to the given graph. The edges are AB, BC, CD, DA, AC . The degrees are

$$\deg(A) = 3, \quad \deg(B) = 2, \quad \deg(C) = 3, \quad \deg(D) = 2.$$

Here A and C have odd degree. **Conclusion.** Since not all vertices are even, the graph has **no Euler circuit**. However, it has exactly two odd-degree vertices (A and C), so an **Euler path** does exist, running from A to C :

$$A \rightarrow D \rightarrow C \rightarrow B \rightarrow A \rightarrow C \quad (\text{edges } AD, DC, CB, BA, AC).$$

Final answer. No Euler circuit (because A and C are odd). But an Euler path exists: $A \rightarrow D \rightarrow C \rightarrow B \rightarrow A \rightarrow C$.

Q17. (7 marks) Prove that in a simple graph the number of vertices of odd degree is always even. **Solution — full explanation.**

What the question asks: prove that odd-degree vertices must come in pairs. **Tool.** The Handshaking theorem: $\sum_{v \in V} \deg(v) = 2e$. **Proof.** Split the vertices into those of even degree (set V_1) and those of odd degree (set V_2). Then

$$\underbrace{\sum_{v \in V_1} \deg(v)}_{\text{even (sum of evens)}} + \sum_{v \in V_2} \deg(v) = 2e.$$

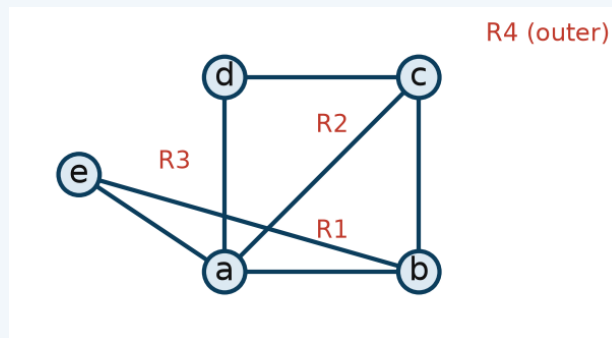
The right-hand side $2e$ is even, and the first sum on the left is even. Therefore the second sum,

$\sum_{v \in V_2} \deg(v)$, must also be even. **Finish.** But $\sum_{v \in V_2} \deg(v)$ is a sum of *odd* numbers. A sum of odd numbers is even only when there is an **even number** of them. Hence the number of vertices of odd degree is even. ■ **Final answer.** The number of odd-degree vertices in a simple graph is always even.

Q18. (7 marks) State and prove Euler's formula for a connected planar simple graph. **Solution — full explanation.**

What the question asks: state the formula linking vertices, edges and regions, and prove it. **Statement.** For a connected planar graph with v vertices, e edges and r regions (faces),

$$v - e + r = 2.$$



Proof (by induction on the number of edges).

- **Base case:** a single edge has $v = 2$, $e = 1$, $r = 1$, and $v - e + r = 2 - 1 + 1 = 2$. ✓
- **Inductive step:** assume the formula holds for a graph G_n , and add one edge to get G_{n+1} . There are two cases.
 - **Case 1** (new edge joins two existing vertices, splitting a region): r increases by 1, e increases by 1, v unchanged. Then $v - e + r$ changes by $-1 + 1 = 0$, so it stays 2.
 - **Case 2** (new edge adds a new vertex, a pendant edge): e increases by 1, v increases by 1, r unchanged. Then $v - e + r$ changes by $+1 - 1 = 0$, so it stays 2.

In both cases the value $v - e + r$ is preserved at 2, so the formula holds for every connected planar graph. ■ **Final answer.** Euler's formula: $v - e + r = 2$ for a connected planar graph.

Q19. (7 marks) Prove that every connected simple planar graph has a vertex of degree at most 5. **Solution — full explanation.**

What the question asks: show some vertex must have degree ≤ 5 . **Step 1 — a useful bound.** For a connected simple planar graph with $v \geq 3$ vertices, the boundary of every region has at least 3 edges, so $2e \geq 3r$. Combining with Euler's formula $r = e - v + 2$ gives

$$3(e - v + 2) \leq 2e \Rightarrow e \leq 3v - 6 \Rightarrow 2e \leq 6v - 12.$$

Step 2 — suppose the opposite. Assume, for contradiction, that *every* vertex has degree at least 6. Then by the Handshaking theorem,

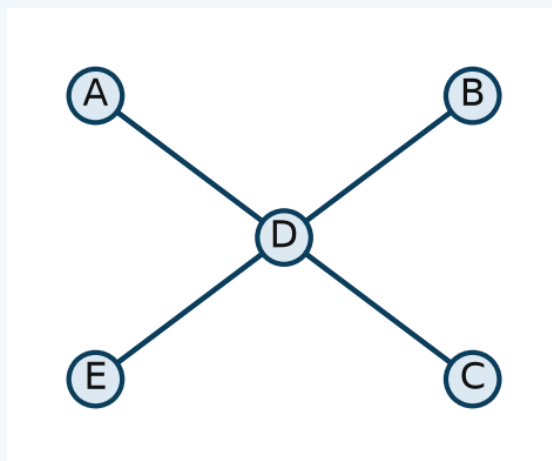
$$2e = \sum_v \deg(v) \geq 6v.$$

Step 3 — contradiction. But Step 1 gave $2e \leq 6v - 12$, which is less than $6v$. The two cannot both hold. **Conclusion.** Our assumption was wrong, so at least one vertex has degree ≤ 5 . ■ **Final answer.** Every connected simple planar graph has a vertex of degree at most 5.

Q20. (7 marks) Use Fleury's algorithm to construct an Euler circuit in the following graph.

Solution — full explanation.

What the question asks: trace an Euler circuit (covering every edge once) using Fleury's algorithm. **The algorithm and why.** The key rule is *do not burn your bridges*: at each step move along any edge that is **not a bridge** (a bridge is an edge whose removal disconnects the graph), unless no other edge is available. Start anywhere if all degrees are even; delete each edge after using it.



Step-by-step. All vertices have even degree ($\deg(D) = 4$, and A, B, C, E each have degree 2), so an Euler circuit exists; we may start at A .

- Start at A ; take edge AD (the only edge at A).
- At D we may take DB , DC or DE . Taking DC would disconnect the graph (it is a bridge), so we take DE .
- At E the only option is EB ; then from B we take BD .
- At D the remaining edges are DC , then CA , closing the circuit.

Circuit. $A \rightarrow D \rightarrow E \rightarrow B \rightarrow D \rightarrow C \rightarrow A$. **Final answer.** Euler circuit: $A - D - E - B - D - C - A$.

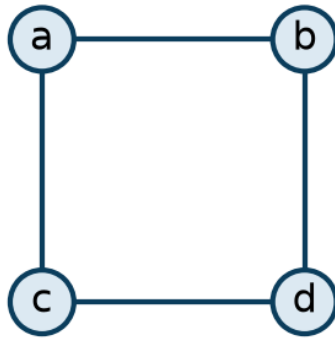
Q21. (7 marks) Construct the adjacency matrix of the following graph. How many paths of length four are there from a to d ? **Solution — full explanation.**

What the question asks: write the adjacency matrix, then count walks of length 4 from a to d .

Adjacency matrix. Order the vertices a, b, c, d . Entry (i, j) is 1 if the two vertices are joined by an edge, else 0:

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

(The graph is the 4-cycle $abdc a$.) The matrix is symmetric, as always for an undirected graph.



Key theorem and why. The number of paths of length k from v_i to v_j equals the (i, j) entry of A^k . So we need the (a, d) entry of A^4 . **Compute.** Evaluating A^4 gives

$$A^4 = \begin{pmatrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{pmatrix}.$$

Hence the (a, d) entry is 8. **The eight paths.**

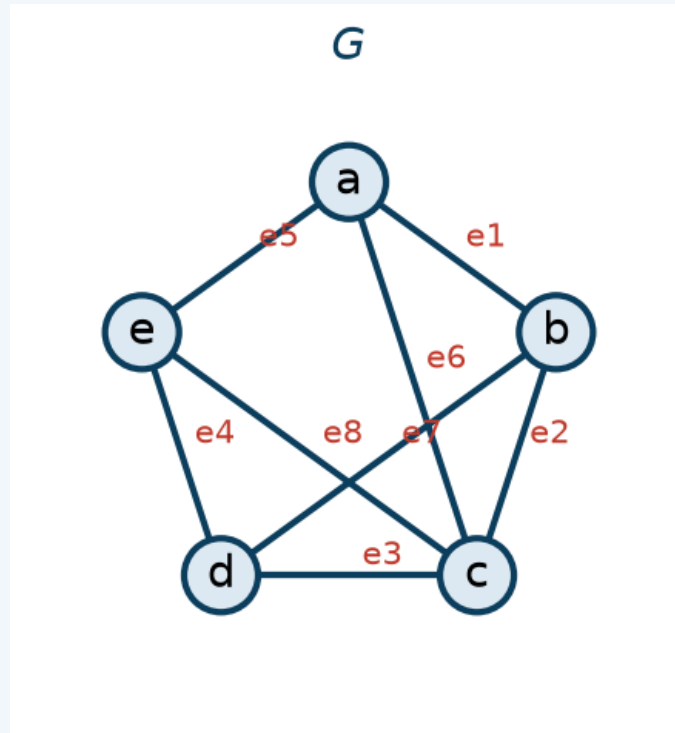
a, b, a, b, d ; a, b, a, c, d ; a, b, d, b, d ; a, b, d, c, d ; a, c, a, b, d ; a, c, a, c, d ; a, c, d, b, d ; a, c, d, c, d .

Final answer. The adjacency matrix is as above, and there are 8 paths of length four from a to d .

Q22. (7 marks) Define the adjacency matrix and the incidence matrix. Find both for the following graph G . **Solution — full explanation.**

What the question asks: define the two matrix representations, then build them for the given graph. **Definitions.** Let G have vertices $v_1, \dots, v_n$ and edges $e_1, \dots, e_m$.

- **Adjacency matrix** $A = [a_{ij}]$ ($n \times n$): $a_{ij} = 1$ if $\{v_i, v_j\}$ is an edge, else 0.
- **Incidence matrix** $M = [m_{ij}]$ ($n \times m$): $m_{ij} = 1$ if vertex v_i is incident with edge e_j , else 0 (a loop contributes 2).



Given graph. Vertices a, b, c, d, e ; edges $e_1 = ab$, $e_2 = bc$, $e_3 = cd$, $e_4 = de$, $e_5 = ea$, $e_6 = ac$, $e_7 = bd$, $e_8 = ce$. **Adjacency matrix** (order a, b, c, d, e):

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

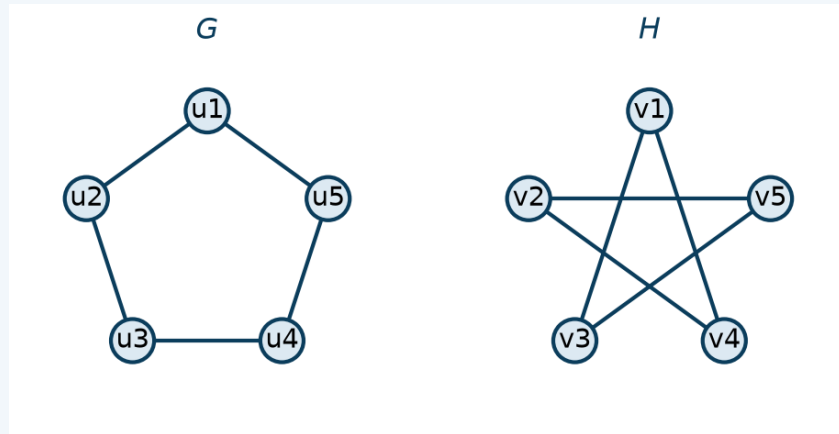
Incidence matrix (rows $a, \dots, e$; columns $e_1, \dots, e_8$):

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

Check. Each column of M has exactly two 1s (each edge touches two vertices); each row-sum of M equals the degree of that vertex (e.g. row c sums to 4). **Final answer.** A and M are as shown above.

Q23. (7 marks) Test whether the graphs G and H given below are isomorphic or not. **Solution** — full explanation.

What the question asks: decide whether the two graphs are “the same graph” up to renaming of vertices. **Definition.** Two graphs are **isomorphic** if there is a bijection f from the vertices of G to those of H such that two vertices are adjacent in G exactly when their images are adjacent in H . **Method and why.** First compare *graph invariants* (properties preserved by isomorphism): number of vertices, number of edges, and the degree sequence. If they match, we try to find an explicit mapping.



Compare invariants.

- Vertices: $|V(G)| = |V(H)| = 5$. ✓
- Edges: $|E(G)| = |E(H)| = 5$. ✓
- Degree sequence: every vertex has degree 2 in both, giving $(2, 2, 2, 2, 2)$ for each. ✓

Both are in fact 5-cycles (G is a pentagon; H is the pentagram, whose edges $v_1v_3, v_3v_5, v_5v_2, v_2v_4, v_4v_1$ also form a 5-cycle). **Explicit isomorphism.** Map

$$f(u_1) = v_1, \quad f(u_2) = v_3, \quad f(u_3) = v_5, \quad f(u_4) = v_2, \quad f(u_5) = v_4.$$

Check the edges: $u_1u_2 \rightarrow v_1v_3$ ✓, $u_2u_3 \rightarrow v_3v_5$ ✓, $u_3u_4 \rightarrow v_5v_2$ ✓, $u_4u_5 \rightarrow v_2v_4$ ✓, $u_5u_1 \rightarrow v_4v_1$ ✓. Every edge of G maps to an edge of H . **Final answer.** G and H are **isomorphic** — both are 5-cycles.